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Jeff Shen

: github.com/al-jshen
in: linkedin.com/in/jeff-shen

Education

Ph.D., Princeton University. Astrophysical Sciences. 2022 - 2027

H.B.Sc., University of Toronto. Astronomy & Astrophysics and Statistics. 2018 - 2022

Publications

8. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, et. al (2023). Stellar parameters for 220m stars from Gaia DR3 BP/RP spectra. In prep.

7. **Jeff Shen**, Peter Melchior (2023). Multiscale Feature Attribution for Outliers. Accepted to NeurIPS Workshop on Machine Learning and the Physical Sciences on 27 Oct 2023. arXiv: 2310.20012.

6. **Jeff Shen**, Joshua S. Speagle, J. Ted Mackereth, Yuan-Sen Ting, Jo Bovy (2023). Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution. Accepted to ApJ on 17 Oct 2023. arXiv: 2305.15634.

5. Yan Liang, Peter Melchior, ChangHoon Hanh, **Jeff Shen**, Andy Goulding, Charlotte Ward (2023). Outlier Detection in the DESI Bright Galaxy Survey. ApJL 956 L6. arXiv: 2307.07664.

4. Seery Chen, Deborah M. Lokhorst, **Jeff Shen**, Imad Pasha, Evgeni I. Malakhov, Roberto Abraham, Pieter van Dokkum (2022). The Dragonfly Spectral Line Mapper: design and first light. Proc. SPIE 12182, Ground-based and Airborne Telescopes IX, 121824E. arXiv: 2209.07489.

3. Deborah M. Lokhorst, Seery Chen, Imad Pasha, **Jeff Shen**, Evgeni I. Malakhov, Roberto G. Abraham, Pieter van Dokkum. The pathfinder Dragonfly Spectral Line Mapper: pushing the limits for ultra-low surface brightness spectroscopy. Proc. SPIE 12182, Ground-based and Airborne Telescopes IX, 121821T. arXiv: 2209.07487

2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han (2022). The Mass of the Milky Way from the H3 Survey. ApJ 925 1. arXiv: 2111.09327.

1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard (2021). Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$. ApJ 917 79. arXiv: 2105.11572.

Honours and Awards

2022 Jan	**Astrostatistics Student Paper Competition Finalist** Astrostatistics Interest Group of the American Statistical Association
2021 - 2022	**Dean's List Scholar** University of Toronto
2021 May - 2021 Aug	**Undergraduate Summer Research Fellowship, ($8,396 CAD)** Canadian Institute for Theoretical Astrophysics, University of Toronto
2020 May - 2020 Aug	**Summer Undergraduate Research Program Award ($9,500 CAD)** Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Talks

2022 August **JSM 2022, Washington, D.C.**
Contributed talk: “The Mass of the Milky Way from the H3 Survey.”

2021 August **SDSS 2021 Collaboration Meeting, JHU/Online**
Lightning talk: “Predicting the ages of stars with machine learning.”

2021 May **Stellar Stats Workshop, UofT/Online**
Lightning talk: “Estimating the mass distribution of the Milky Way with Bayesian multilevel models.”

2021 May **Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online**
Lightning talk: “Molecular gas in a gravitationally lensed galaxy group at z = 2.9.”

2021 Apr **Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online**
“Molecular gas in gravitationally lensed galaxies at z = 2.9.”

2020 Oct **REQUIEM Galaxies Telecon, Online**
“Molecular gas in gravitationally lensed galaxies at z = 2.9.”

Posters

2023 Jan **241st Meeting of the American Astronomical Society, Seattle**
“Stellar parameters for 220m stars from *Gaia* DR3 BP/RP spectra.”

2021 May **Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online**
“Molecular gas in a gravitationally lensed galaxy group at z = 2.9.”

2020 Aug **Summer Undergraduate Research Program Poster Session, UofT/Online**
“Molecular gas in a gravitationally lensed galaxy group at z = 2.9.” Awarded best poster.

Outreach

2019 - 2021 **Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast
Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - end of show.

2015 - 2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for national programming contest.

Media

2021 Jul **National Ratio Astronomy Organization (NRAO) eNews**
Digital newsletter feature: “Molecular gas in high redshift galaxies”. https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas

2020 Jul **SURP Student of the Week Feature**
Interview: https://www.dunlap.utoronto.ca/surp-sow/sow12020/

Technical Skills

JAX, Stan, Pytorch, SQL, R, C++, Rust, BLAS/LAPACK, LaTeX, regex, bash, git, Docker, Javascript, React